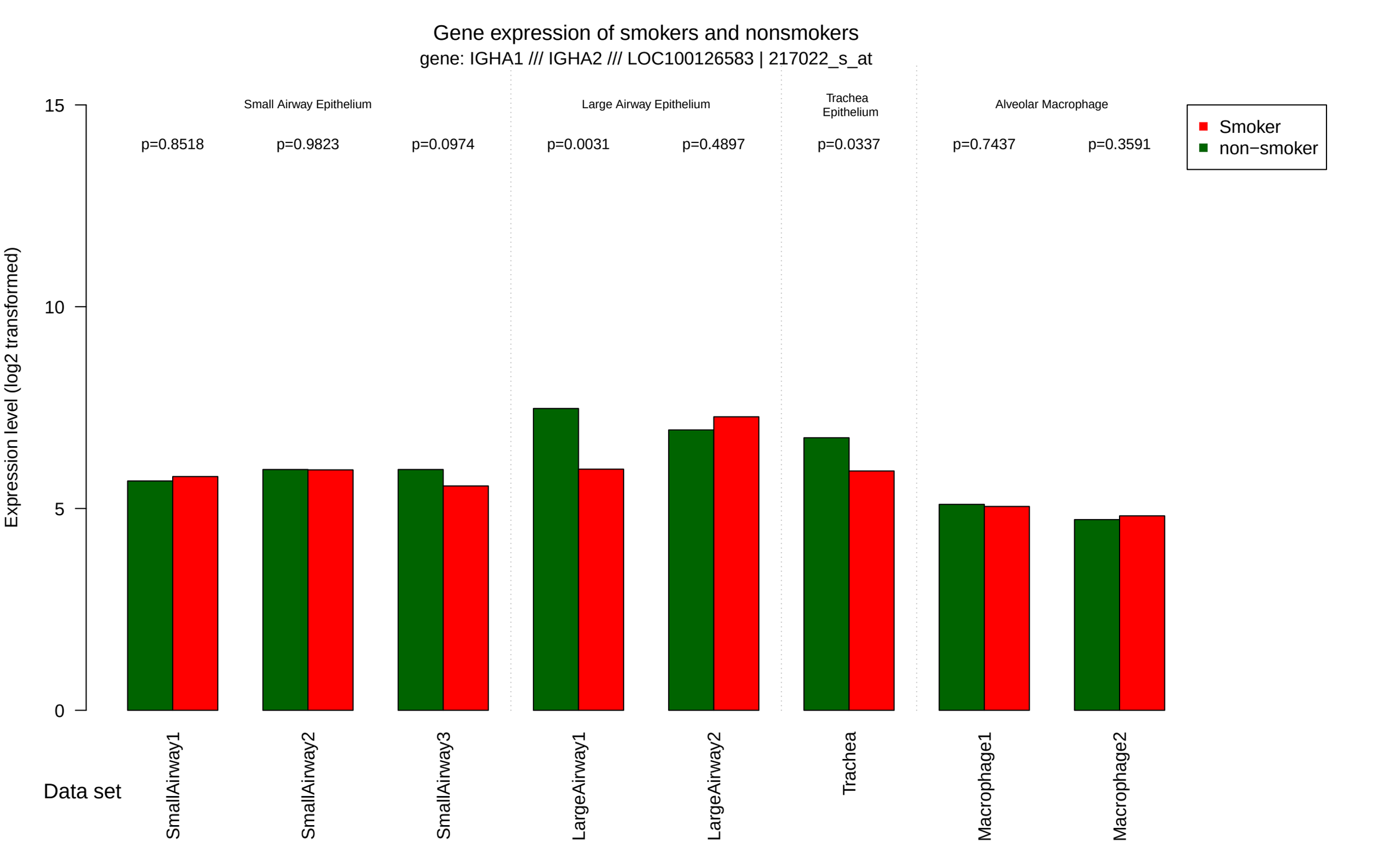
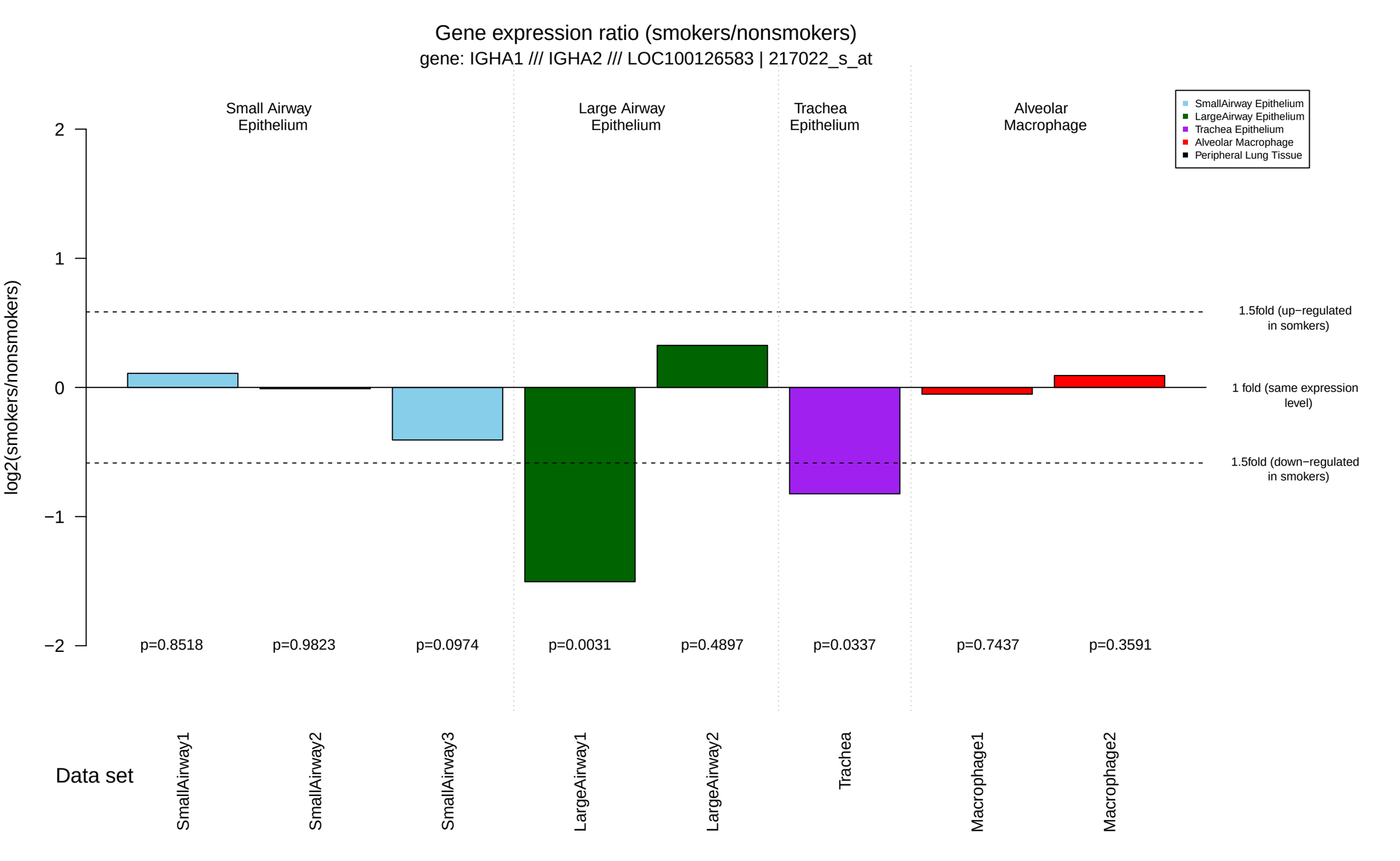
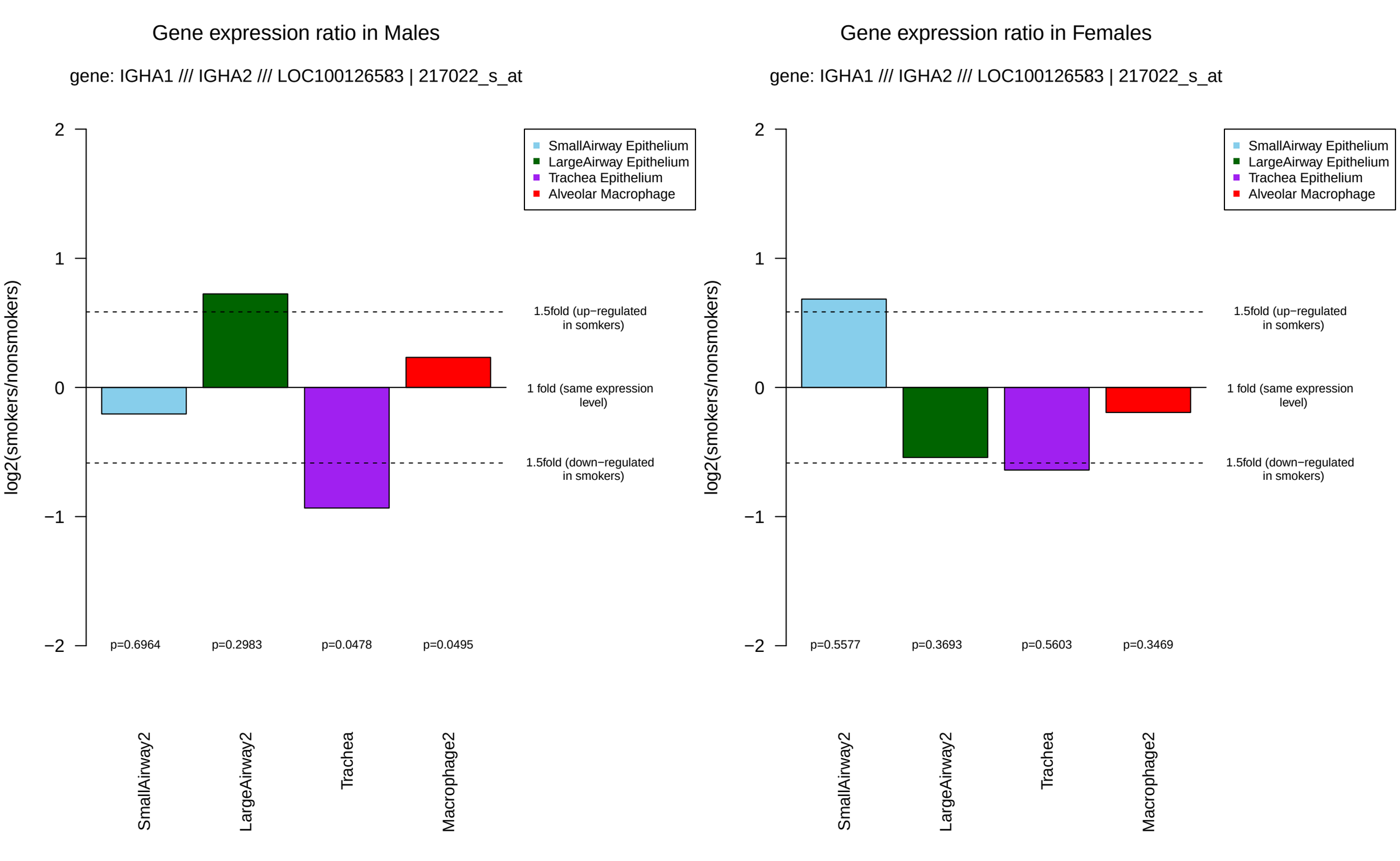
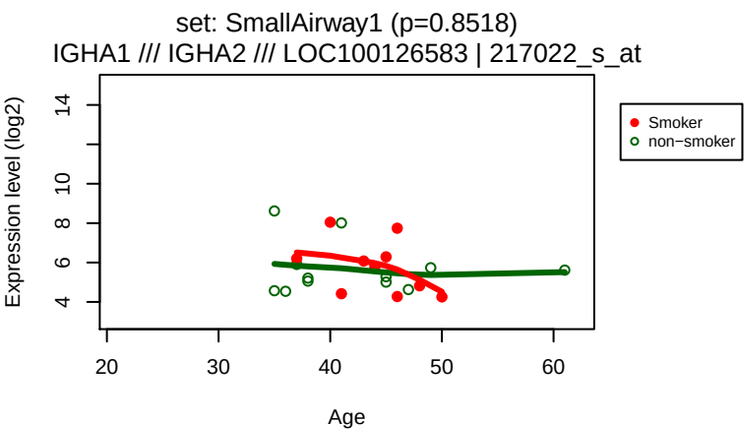
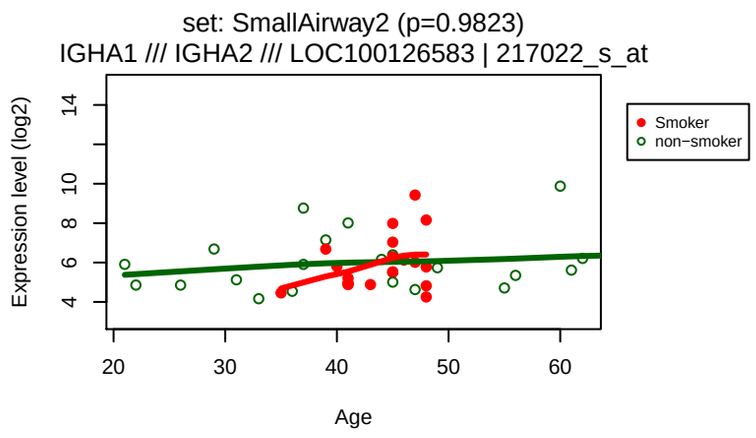
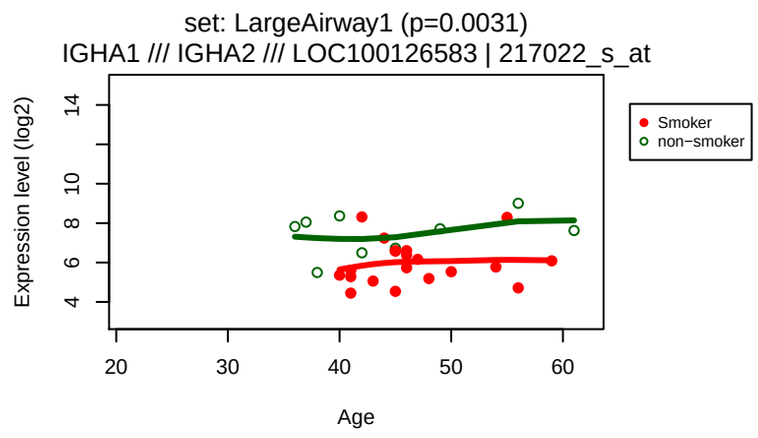

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHA2 /// LOC100126583 | 217022_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8518
p=0.9823
p=0.0974
p=0.0031
p=0.4897
p=0.0337
p=0.7437
p=0.3591
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

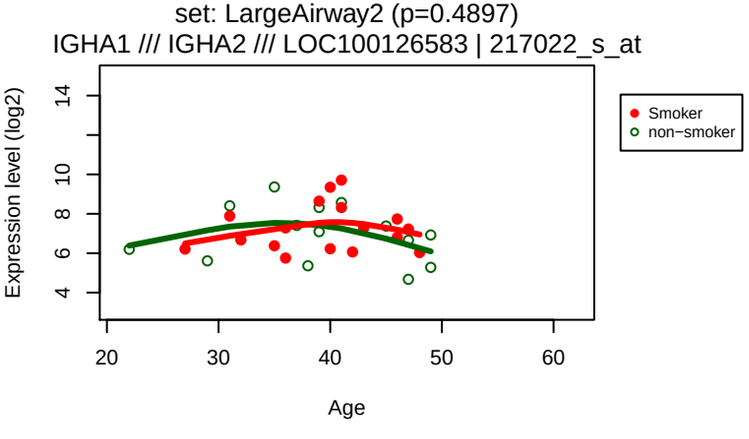

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHA2 /// LOC100126583 | 217022_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8518
p=0.9823
p=0.0974
p=0.0031
p=0.4897
p=0.0337
p=0.7437
p=0.3591
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

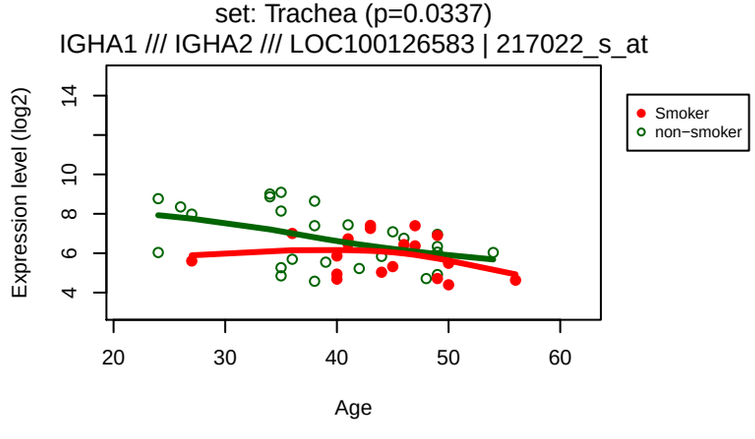

Gene expression ratio in Males
gene: IGHA1 /// IGHA2 /// LOC100126583 | 217022_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6964
p=0.2983
p=0.0478
p=0.0495
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHA1 /// IGHA2 /// LOC100126583 | 217022_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5577
p=0.3693
p=0.5603
p=0.3469
SmallAirway2
LargeAirway2
Trachea
Macrophage2

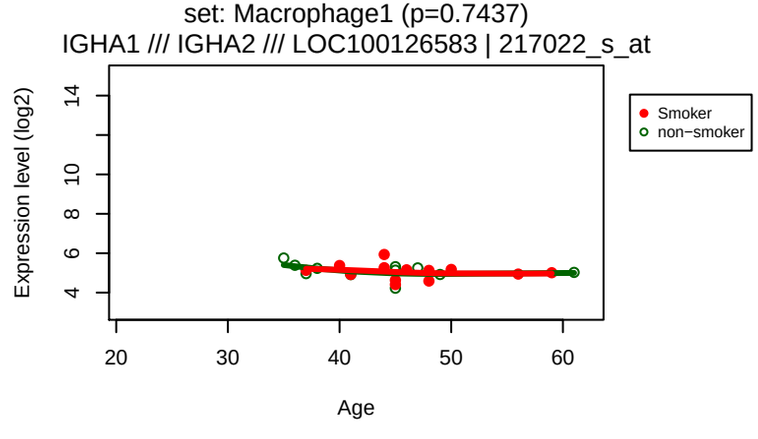

set: SmallAirway1 (p=0.8518)
IGHA1 /// IGHA2 /// LOC100126583 | 217022_s_at
Expression level (log2)
Age
Smoker
non−smoker

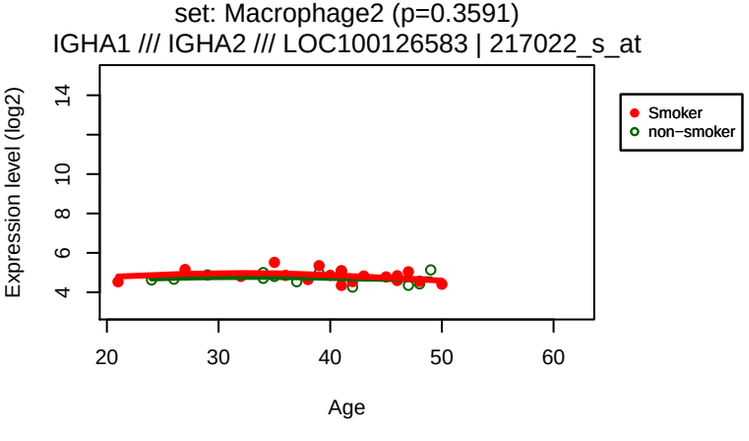

set: SmallAirway2 (p=0.9823)
IGHA1 /// IGHA2 /// LOC100126583 | 217022_s_at
Expression level (log2)
Age
Smoker
non−smoker

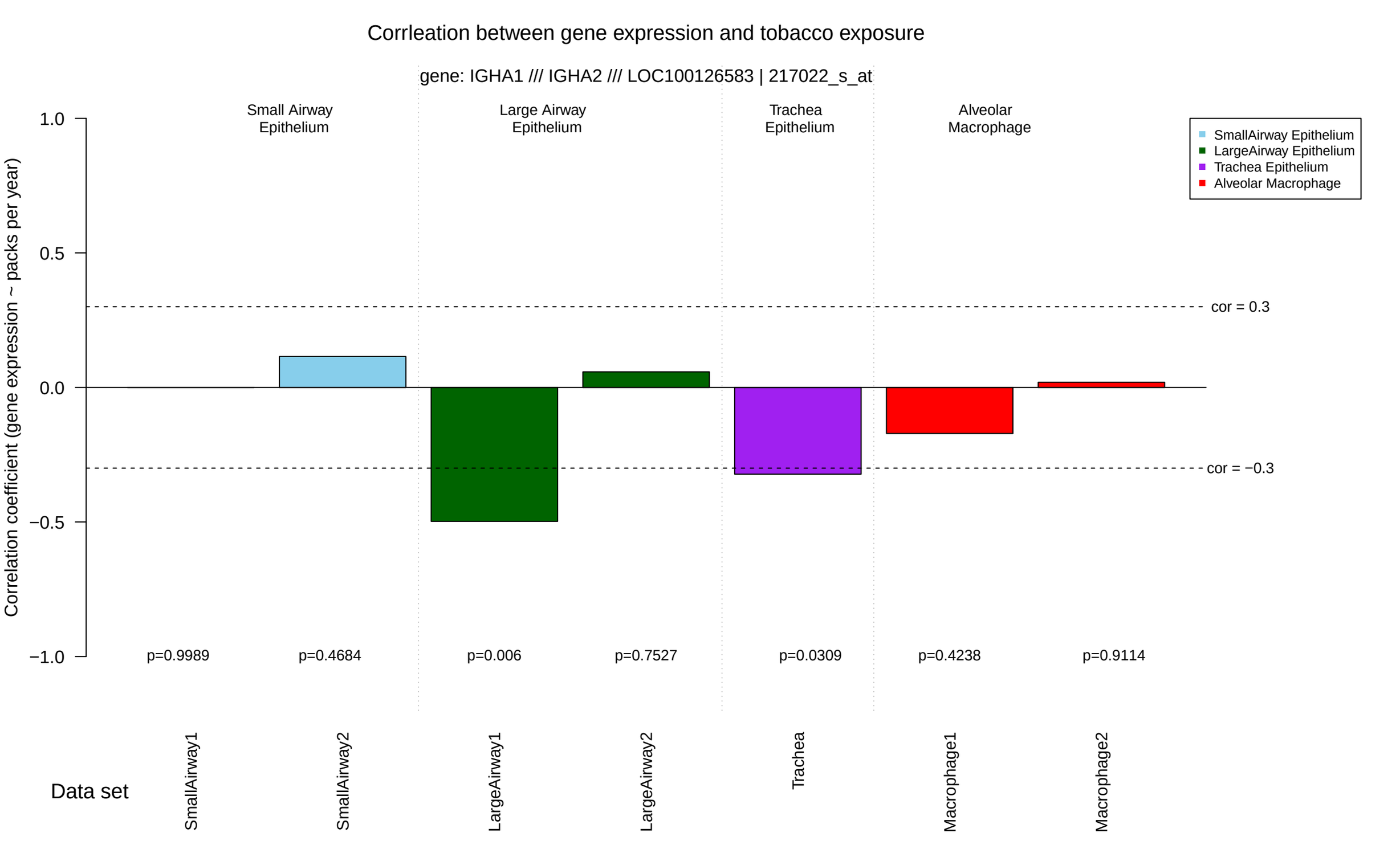

set: LargeAirway1 (p=0.0031)
IGHA1 /// IGHA2 /// LOC100126583 | 217022_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4897)
IGHA1 /// IGHA2 /// LOC100126583 | 217022_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0337)
IGHA1 /// IGHA2 /// LOC100126583 | 217022_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7437)
IGHA1 /// IGHA2 /// LOC100126583 | 217022_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3591)
IGHA1 /// IGHA2 /// LOC100126583 | 217022_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHA2 /// LOC100126583 | 217022_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9989
p=0.4684
p=0.006
p=0.7527
p=0.0309
p=0.4238
p=0.9114
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2